



Real Problems. Real Solutions. Real Impact

Smart Wheezing Pre-Detection and Alert System

Mapped SDG:3 Good Health and Well-Being

The project focuses on early detection and prevention of respiratory emergencies like wheezing and asthma attacks, improving health outcomes and reducing preventable hospitalizations.

Project Description:

Wheezing and asthma patients often experience sudden breathing difficulties without early warning. Existing devices detect wheezing only after an attack has started.

This project proposes a smart, affordable system that continuously monitors breathing patterns and environmental factors to predict wheezing attacks before they occur and sends timely alerts to patients and caregivers, enabling early intervention and reducing health risks.

Project Relevant Photo



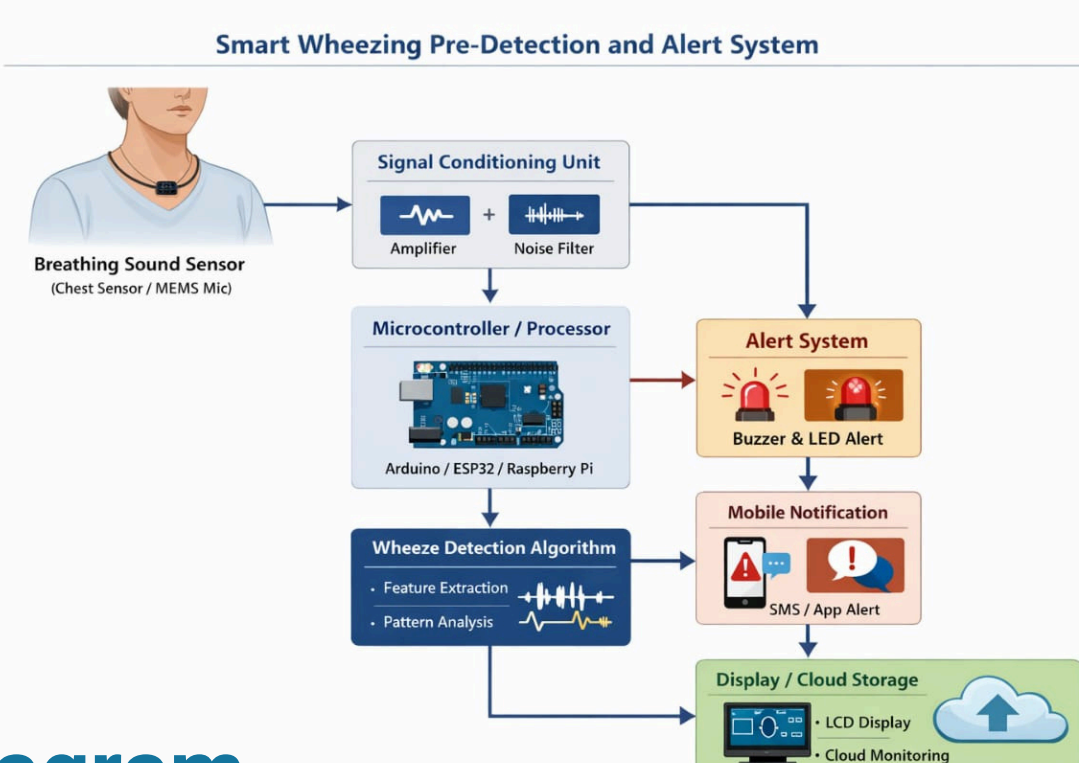
Objective:

- To continuously monitor breathing patterns of wheezing/asthma patients
- To detect early abnormalities before a wheezing attack begins
- To provide instant alerts to patients and caregivers
- To develop a low-cost and user-friendly health monitoring solution

Expected Outcome:

- Early warning of wheezing or asthma attacks
- Reduced severity of respiratory emergencies
- Improved patient safety and confidence
- Lower healthcare costs due to fewer hospitalizations
- Increased awareness of respiratory health and pollution impact

Block Diagram



Expected Beneficiaries:

- Asthma and wheezing patients
- Children and elderly people with respiratory problems
- Caregivers and family members
- Hospitals and healthcare professionals
- People living in high-pollution or low-resource areas

